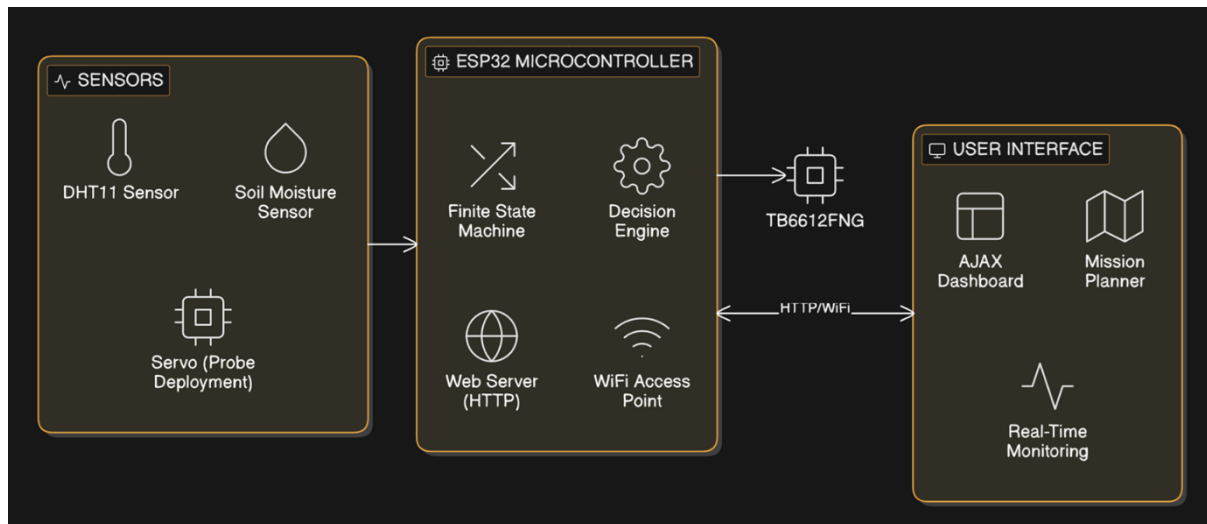


REVIEW DOCUMENT

ESP32-Based Smart Agriculture Rover for Multi-Point Monitoring and Decision Support

1. SYSTEM ARCHITECTURE

1.1 Conceptual Block Diagram



2. COMPONENT ANALYSIS

2.1 Microcontroller (ESP32)

Working Principle

ESP32 is a 32-bit microcontroller with integrated WiFi, used to process sensor data, execute control logic, and host a web server.

Interfacing

- ADC: Soil sensor (GPIO 34)
- Digital GPIO: DHT11 (GPIO 4)
- PWM: Servo (GPIO 5)
- GPIO: Motor driver control

On-Chip Peripherals Used

- ADC
- GPIO
- Timers (millis-based scheduling)

- WiFi module
- Interrupt system (conceptually supported)

Programming Level

- API level (Arduino framework)

2.2 Soil Moisture Sensor

Working Principle

Provides analog voltage proportional to soil moisture.

Interfacing

- 1 ADC pin (GPIO 34)

Calibration

- Dry soil value recorded
- Wet soil value recorded
- Threshold derived experimentally

Strategy

- Polling at periodic intervals

2.3 DHT11 Sensor

Working Principle

Digital sensor providing temperature and humidity using a timing-based protocol.

Interfacing

- Single GPIO (GPIO 4)

Library

- DHT.h

Strategy

- Periodic polling

2.4 Servo Motor

Working Principle

Uses PWM signal to control angular position.

Interfacing

- PWM pin (GPIO 5)

Function

- Deploys soil sensor for measurement

2.5 Motor Driver (TB6612FNG)

Working Principle

Dual H-bridge motor driver enabling bidirectional control.

Interfacing

- IN1, IN2, IN3, IN4 → Direction control
- PWMA, PWMB → Speed control
- STBY → Enable

2.6 Communication Module

Protocol

- WiFi (Access Point Mode)
- HTTP
- AJAX polling

Function

- Enables real-time communication with web dashboard

3. PROGRAMMING LOGIC

3.1 Sequential Model (Pseudocode)

Initialize system

Loop:

 Handle web requests

 If mission active:

 Execute FSM

 Else:

 Wait for user input

3.2 Decision Logic

IF soil_moisture < threshold:

 Suggest irrigation

ELSE IF temperature > threshold:

 Suggest temperature alert

ELSE IF humidity < threshold:

 Suggest humidity warning

ELSE:

 System optimal

3.3 Polling vs Interrupt

Component Strategy

Soil Sensor Polling

DHT11 Polling

Web Server Event-driven

FSM Timer-based

4. FINITE STATE MACHINE (FSM)

4.1 States

State	Description
IDLE	Waiting for command
RUNNING	Executing movement
PROBING	Sensor data collection
DONE	Mission complete

4.2 State Transition Table

Current State	Condition	Next State
IDLE	Mission start	RUNNING
RUNNING	Time elapsed	PROBING / NEXT
PROBING	Completed	RUNNING
RUNNING	Final step	DONE

4.3 FSM Characteristics

- Non-blocking execution using millis()
- Real-time responsiveness
- Event-driven transitions

5. CONCURRENT PROGRAMMING MODEL

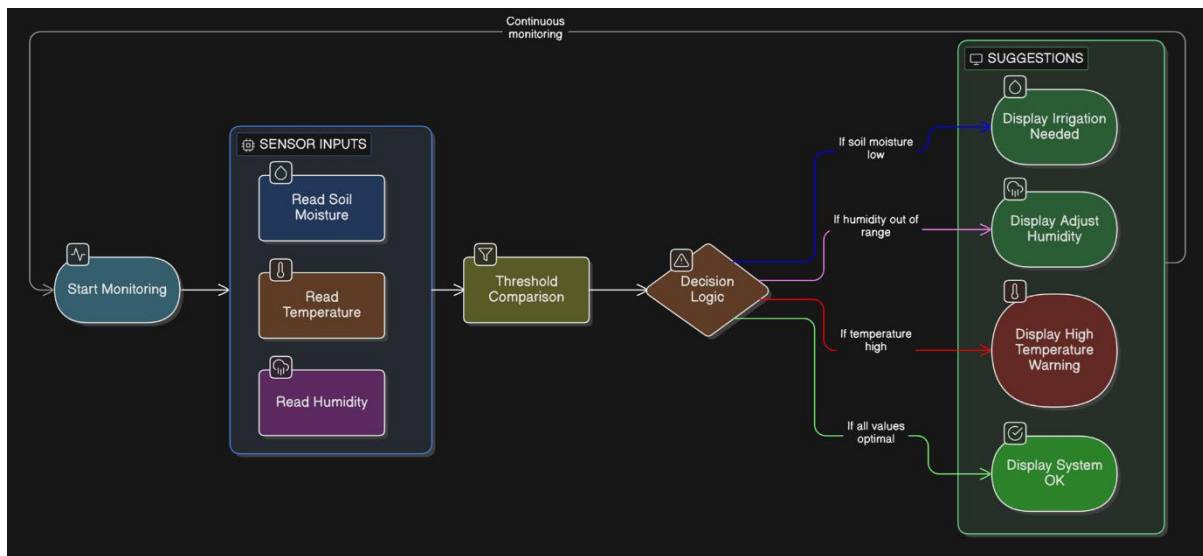
The system demonstrates logical concurrency:

- Web server handling
- Sensor acquisition
- FSM execution

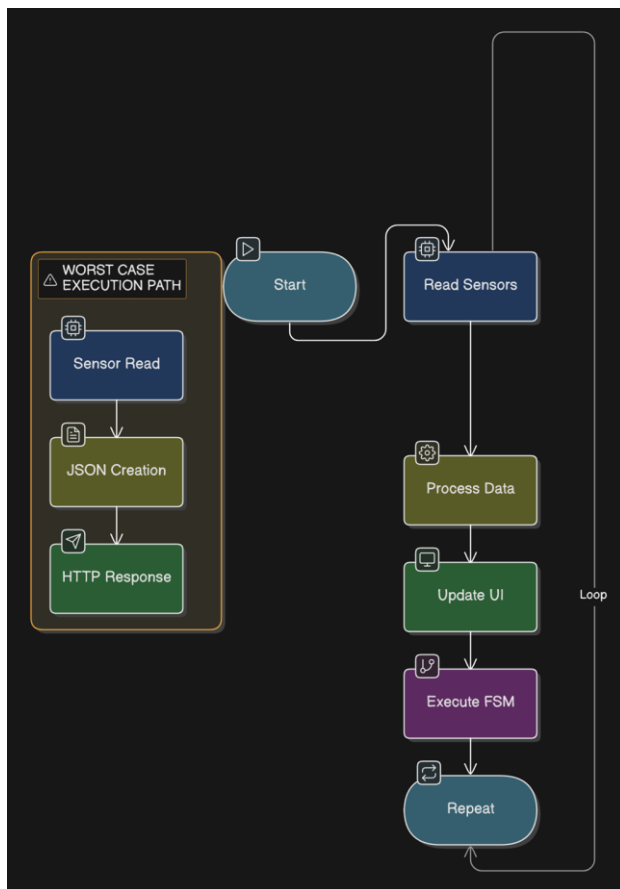
Implemented using:

- Cooperative multitasking (loop-based execution)

6. DATA FLOW GRAPH (DFG)



7. CONTROL DATA FLOW GRAPH (CDFG)



8. CODE OPTIMIZATION

Time Optimization

- Non-blocking design using millis()
- FSM implementation

Space Optimization

- Static arrays for waypoints
- Minimal dynamic memory usage

Energy Optimization

- Controlled execution
- Reduced unnecessary operations

Compiler Optimization

- Default Arduino optimization flags (-Os)

9. REAL-TIME SYSTEM ANALYSIS

Task Classification

Task	Type
FSM execution	Soft real-time
Sensor reading	Soft real-time
Web server	Non real-time

Events:

Event	Type
Sensor read	Periodic
HTTP request	Asynchronous
FSM transition	Event-driven

Timing Parameters

- Sensor update rate: 1 Hz
- Decision latency: < 1 second

Scheduling

- Event-driven + periodic scheduling
- Single processor system

Threads

Implemented conceptually via:

- Main loop execution
- Non-blocking tasks

10. IMPLEMENTATION STATUS

Motor control

Sensor integration

FSM-based navigation

Web dashboard

Decision logic

11. ANALYSIS

11.1 Dataset

- Real-time sensor data
- No external dataset required

11.2 Algorithm

- Threshold-based decision algorithm
- Multi-parameter evaluation

11.3 Execution Platform

- Edge computing (ESP32)

11.4 User Interface

- Web-based dashboard
- AJAX-based real-time updates
- Mission planner

11.5 Test Cases

Test Case	Input	Output
Dry soil	Low value	Water required
High temperature	> threshold	Warning
Normal	Optimal	System OK

12. PERFORMANCE METRICS

Quantitative

Metric	Value
Response time	< 1 sec
Update frequency	1 Hz
Memory usage	Low
Power consumption	Moderate

Qualitative

- User-friendly interface
- Real-time monitoring
- Scalable architecture

13. SENSOR VALIDATION

- Calibration using known dry/wet values
- Multiple readings averaged
- Error handling via thresholding

14. COMMUNICATION EFFICIENCY

- Packet delivery: High (local AP)
- Packet loss: Minimal
- Protocol: HTTP over WiFi

15. SPECIAL PURPOSE SYSTEM ANALYSIS (CO2)

The system is a mobile embedded agricultural monitoring system designed for:

- Low cost
- Real-time operation
- Limited power

16. MCU SELECTION JUSTIFICATION (CO1)

ESP32 selected due to:

- Integrated WiFi
- ADC capability
- Low cost
- Sufficient processing power

17. NETWORKING PROTOCOL (CO4)

- WiFi (IEEE 802.11)
- HTTP protocol
- AJAX for real-time updates

18. HARDWARE PROTOTYPE

- ESP32 development board
- Soil sensor
- DHT11
- Servo motor
- Motor driver + rover chassis

19. COMPARATIVE ANALYSIS

Feature	Existing Systems	Proposed System
Monitoring	Yes	Yes
Mobility	No	Yes
Decision Logic	Limited	Yes
Cost	High	Low
Edge Processing	Partial	Full

20. FUTURE WORK

The current system focuses on monitoring and decision support. The following enhancements are proposed:

1. Integration of **automatic irrigation system using relay-controlled water pumps**
2. Addition of **pesticide and nutrient spraying mechanisms** based on environmental conditions
3. Implementation of **advanced prediction models (e.g., LSTM)** for rainfall and irrigation scheduling
4. Integration of **camera-based crop monitoring**
5. Multi-rover coordination for large-scale farms

21. CONCLUSION

The project successfully demonstrates a mobile embedded system for agricultural monitoring and decision support. The integration of sensors, FSM-based control, and real-time web interface ensures efficient operation. The system is scalable, cost-effective, and suitable for precision agriculture applications.